



Beyond Similarity Scoring: Detecting Entailment and Contradiction in Multilingual and Multimodal Contexts

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Abstract

Natural Language Inference (NLI) determines whether a hypothesis entails, contradicts, or is neutral with respect to a premise. While text-based NLI is well-studied, its multimodal and multilingual extension remains underexplored. This paper introduces a multilingual, multimodal NLI framework classifying entailment, contradiction, and neutrality across text-text, text-speech, speech-text, and speech-speech pairs in same- and cross-lingual settings. A key motivation is improving translation assessment, where similarity-based approaches may miss contradictions. The framework complements evaluation methods and helps identify inconsistencies by detecting entailment and contradiction alongside semantic similarity. It also extends text-based datasets with speech-text and speech-speech pairs for multilingual multimodal inference.¹ Experiments show the model outperforms BLASER in distinguishing entailment from non-entailment, achieving F1 gains of 0.19 in speech-speech and 0.13 in speech-text.

Index Terms: Multimodal NLI, Speech Translation Evaluation

1. Introduction

NLI determines the logical relationship between a premise and a hypothesis. The premise is a given statement, and the hypothesis is evaluated against it. The relationship can be entailment (the hypothesis follows from the premise), contradiction (it conflicts with the premise), or neutral (it is unrelated or ambiguous) [1, 2]. NLI is used in tasks like information extraction and virtual assistants [3, 4]. Multilingual and multimodal NLI extends this capability to spoken and translated sentences, where implicit meaning shifts and contradictions can impact evaluation. In translation, bidirectional entailment [5] ensures that the translated and original sentences support each other and detects translations that contradict the original speech. Beyond translation, NLI plays a crucial role in speech recognition, where phonetic variations and transcription errors can introduce contradictions. This is particularly relevant in morphologically rich languages like Arabic, where minor phonetic changes can alter meaning and impact transcription accuracy.

Early NLI methods used syntactic and lexical features but struggled with semantic relationships [1, 6]. The shift to deep learning, particularly LSTM and attention-based mechanisms, enabled more robust semantic modeling and significantly improved inference performance [7]. Transformers advanced NLI using pre-trained contextualized representations. Models like BERT [8] and RoBERTa [9] improved performance by capturing semantic relationships more effectively. This progress extended to multilingual models, such as mT5 [10] and mGPT

[11]. Specialized models have further addressed challenges in cross-lingual representation learning: No Language Left Behind (NLLB) [12] enhances low-resource language performance, while LaBSE [13] improves multilingual retrieval by producing language-agnostic sentence embeddings. Together, these advancements have broadened the scope of NLI across languages and modalities.

Expanding NLI to speech introduces challenges like accent variations, diverse patterns, and noise. Self-supervised models like Wav2vec 2.0 [14] and HuBERT [15] generate high-dimensional representations that capture phonetic and linguistic features. However, ensuring alignment between speech and text embeddings is critical for multimodal NLI. SONAR [16] refines speech encoders using NLLB, aligning speech outputs with text embeddings at the sentence level. Similarly, SAMU-XLSR [17] leverages LaBSE for cross-lingual speech-text alignment. These models align speech and text in a shared space, improving multimodal inference.

Cosine similarity compares sentence embeddings based on their angular difference [18, 19]. Other metrics include Euclidean distance, dot product similarity, and learned similarity functions [20, 21]. BLASER 2.0 [22, 23], a three-layer dense neural network, computes semantic similarity between speech and text using a neural similarity function. It is trained with supervised learning on cross-lingual semantic textual similarity (XSTS) data [24], which provides human-annotated scores from 1 (unrelated) to 5 (fully equivalent). This training aligns BLASER’s similarity scores with human judgments. In the SeamlessM4T system, BLASER 2.0 ranks and retrieves matching speech and text segments for speech-text alignment and translation evaluation [22].

Although these metrics measure overall similarity, they may assign high scores to sentences with high lexical overlap but opposite meanings. In our own evaluation, BLASER gives a 4.79/5 similarity score to the sentences “The weather is sunny” and “The weather is not sunny” despite their contradiction. Detecting contradiction prevents misleading similarity scores from affecting retrieval and evaluation. Explicit entailment detection filters mismatches, improving retrieval and avoiding contradictory pairs in multilingual and multimodal settings.

This study presents a framework extending NLI to multilingual and multimodal contexts using semantic representations. It detects entailment and contradiction across input types: text-text, text-speech, speech-text, and speech-speech, within same- and cross-lingual settings. Unlike similarity-based methods, it explicitly classifies entailment, contradiction, and neutrality. It incorporates data generation for speech-text and speech-speech NLI pairs, aiding multimodal datasets. This approach applies to translation assessment, speech recognition, and multilingual multimodal retrieval.

¹The dataset we created is available at: <https://github.com/othman-istaiteh/multimodal-nli>

2. Methodology

This NLI framework supports multilingual and multimodal inputs across text-text, text-speech, speech-text, and speech-speech in same- and cross-lingual settings. Text-text spans 15 languages; speech-text and speech-speech include Arabic, English, and French. First, input pairs are encoded into a shared semantic space for consistent cross-lingual and cross-modal representation. This uses a single encoder for speech and text or multiple aligned encoders for a unified representation. A fully connected neural network with a softmax layer classifies relationships as entailment, contradiction, or neutral.

2.1. Feature Combination of Embeddings

The framework applies feature combination techniques from the BLASER architecture to process premise and hypothesis embeddings. After encoding input pairs into a shared semantic space, their embeddings are combined using element-wise subtraction (to highlight differences in meaning), element-wise multiplication (to capture interactions and shared semantic information), and concatenation (to merge the original, subtracted, and multiplied embeddings into a single vector). The resulting feature vector encodes the premise-hypothesis relationship and has a dimensionality four times that of a single input embedding.

2.2. Classification

The classification module uses the combined feature vector to predict the premise-hypothesis relationship. A dense neural network classifier with a softmax layer assigns probabilities to the three NLI classes: entailment, contradiction, and neutral. By directly modeling these relationships, the framework detects entailment and contradiction across different modalities and languages. It also distinguishes contradiction from entailment in cases with high lexical overlap, where similarity-based models like BLASER may struggle.

The classification network handles text-text, text-speech, speech-text, and speech-speech inputs within and across languages. It directly models logical relationships between inputs, providing a key advantage over regression-based similarity models. Figure 1 illustrates the proposed framework, outlining the process from input encoding to classification. The blue blocks represent the original BLASER architecture, while the red blocks indicate the newly introduced components. This capability benefits translation assessment, where entailment complements high semantic scores, and information retrieval, where logical consistency is essential.

3. Experiments

We conducted experiments to evaluate the proposed framework’s ability to classify entailment, contradiction, and neutral relationships across text-text, text-speech, and speech-speech inputs. The primary task was multiclass classification, distinguishing among the three labels. Additionally, we performed binary classification, where entailment was treated as the positive class, and contradiction and neutral were combined into a single negative class for comparison against BLASER. The experiments also examined the impact of generated data and compared performance on synthesized and natural speech.

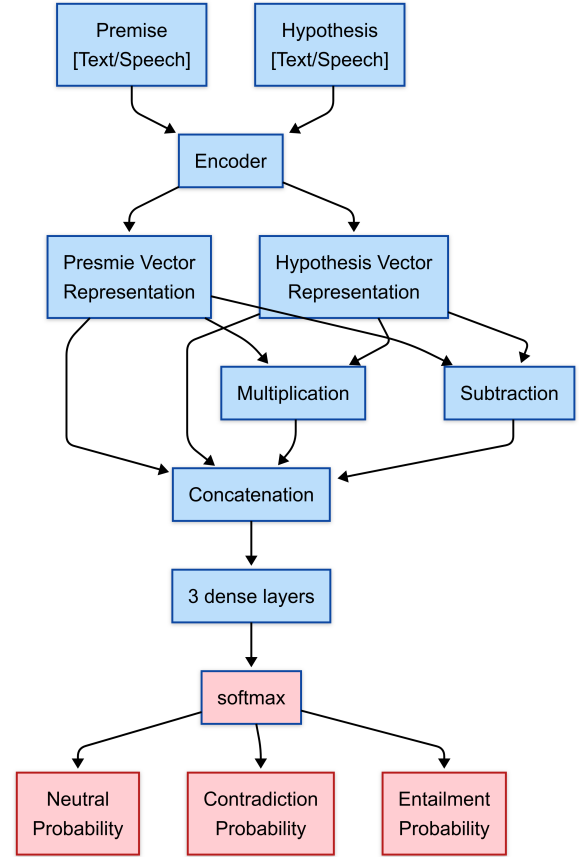


Figure 1: Workflow of the proposed multilingual and multimodal NLI framework.

3.1. Data Preparation

The data preparation consists of two parts. The first utilizes existing datasets, including SNLI [1], XNLI [25], and FLEURS [26]. The second generates additional multimodal and multilingual NLI data using synthetic techniques to extend the datasets into new modalities and languages.

3.1.1. Datasets Utilized

This work uses three publicly available datasets in two categories:

Text Datasets: The SNLI and XNLI corpora are used for NLI tasks. SNLI consists of English premise-hypothesis pairs labeled as entailment, contradiction, or neutral. XNLI extends NLI to 15 languages, with premise and hypothesis in the same language. This work uses the English subset of XNLI for training and the multilingual subset for development and testing.

Speech Dataset: The FLEURS dataset contains aligned text and speech data for 102 languages, with 11,000 samples per language. Each sample includes a sentence translation and its corresponding audio recording. This work focuses on English, French, and Arabic pairs.

3.1.2. Data Generation

We used a two-step process to extend NLI text data to a multimodal context, generating speech-text and speech-speech pairs from existing text-text pairs. Figure 2 illustrates the process.

Text-text Pairs: We fine-tuned the Mistral 7B v3 model [27] on the English portions of SNLI and XNLI. The model generated premise-hypothesis pairs for entailment, contradiction, and neutral labels. Hypotheses for the English text in FLEURS were paired with their Arabic and French translations, forming English-English, English-Arabic, and English-French text-text NLI pairs.

Speech-text Pairs: English NLI-generated pairs were aligned with the corresponding FLEURS speech, forming English-English, English-Arabic, and English-French speech-text NLI pairs.

Speech-speech Pairs: We used the VITS multispeaker [28] implementation of the Coqui open-source framework [29] to generate speech-speech pairs. The 109 available speakers were split into training (77), validation (12), and testing (20), matching FLEURS proportions. Using VITS, we generated speech for NLI-generated pairs and paired it with corresponding FLEURS speech, forming English-English, English-Arabic, and English-French speech-speech pairs. The generated speech was also paired with its corresponding text, adding more speech-text data. Whisper v3 [30] generated transcriptions for the TTS-generated speech, with an average Word Error Rate (WER) of 6.9%. To ensure quality, outputs with WER above 20% were filtered out, removing 884 of 10,677 samples, resulting in a final dataset with an average WER of 3.8%.

The SNLI and XNLI datasets, along with the generated data, are balanced across the three NLI labels: entailment, contradiction, and neutral. In contrast, the control set contains only entailment examples, including text-text and speech-text translations, as well as text paired with synthetic speech from TTS. Table 1 summarizes the size and composition of the dataset.

3.2. Training Setup

The training process integrates SONAR embeddings for a unified representation of text and speech, ensuring alignment across languages and modalities. Each sentence, regardless of modality or language, is encoded into a 1024-dimensional vector. SONAR uses a single text encoder for all languages and separate speech encoders for each language. This alignment makes SONAR suitable for text-text, text-speech, and speech-speech experiments.

The classifier retains BLASER’s three-layer dense neural network but replaces the regression output with a three-node softmax layer to classify sentence pairs as entailment, contradiction, or neutral. The input vector is formed by concatenating the two 1024-dimensional sentence embeddings, along with their element-wise subtraction and multiplication, capturing their relationship. The resulting 4096-dimensional vector is fed into a three-layer dense network with hidden dimensions of 3072, 1536, and 3. The first two layers use the tanh activation function, while the last applies softmax. Training employs a dropout rate of 0.1, a batch size of 2048, a learning rate of 1×10^{-3} , and 1000 epochs, using cross-entropy loss.

Table 1: *Data Summary (Number of Samples)*

Dataset	Train	Valid	Test	Languages
SNLI	548,814	9,840	9,824	EN
XNLI	392,702	37,350	75,150	15 Languages
text-text	40,881	7,827	9,882	EN, AR, FR
speech-text	81,492	14,978	19,145	EN, AR, FR
speech-speech	37,357	7,151	9,263	EN, AR, FR
Control	83,141	13,864	18,542	EN, AR, FR

Table 2: *Binary Classification Results (F1 Micro, Threshold: 3.90 for BLASER)*

Dataset	Baseline (BLASER)	Our Model
Dev All	0.69	0.84
Test All	0.69	0.83
XNLI Test	0.67	0.77
SNLI Test	0.66	0.90
Control Test	0.88	1.00
speech-text Test	0.63	0.82
text-text Test	0.62	0.84
speech-speech Test	0.62	0.81

3.3. Binary Classification: Baseline Comparison

BLASER serves as the baseline due to its semantic scoring across multiple modalities, providing a meaningful comparison for our framework, which explicitly models entailment, contradiction, and neutrality.

To evaluate the framework’s performance and enable a direct comparison with the BLASER model, we created a binary classification task. The original task consists of three classes: entailment, contradiction, and neutral. We converted it to binary classification, treating entailment as positive and merging contradiction and neutral into one negative class. This binary setup allowed us to assess the model’s ability to distinguish entailment from non-entailment relationships.

The BLASER model predicts semantic similarity scores on a scale from 1 to 5. We used F1-Micro, calculating precision and recall globally across all classes without assigning weight to any class. F1-Micro is equivalent to F1-Binary for this binary classification task, as it aggregates precision and recall across both classes. Since BLASER is a regression model and does not require a threshold, we applied a threshold of 3.9, selected to maximize the F1 Score on the development set, enabling a direct comparison with our approach.

Table 2 presents the results, showing that our model consistently outperforms BLASER across all test splits in terms of F1-Micro. These results highlight the framework’s effectiveness in distinguishing entailment from non-entailment with greater precision. Our model offers a complementary translation and transcription evaluation tool, where a high semantic score must align with a clear entailment relationship. This capability is also valuable for information extraction, ensuring that extracted content does not contradict the input sentence.

As a comparison by example, Table 3 demonstrates a case where BLASER assigns a higher similarity score to a contradictory pair than to a correct translation. While both hypotheses are in French, only the first one (‘J’adore le chocolat.’) is a valid translation of ‘I love chocolate,’ yet BLASER assigns a higher similarity score (4.49) to the contradiction. In contrast, our model correctly classifies the first as entailment and the second as a contradiction, demonstrating its ability to detect meaning differences beyond surface similarity.

3.4. Impact of Generated and Synthesized Data on Multiclass Classification

We assessed the impact of generated and synthesized data on multiclass NLI tasks through three experiments. The first experiment used only the XNLI and SNLI datasets as a baseline, achieving F1 Micro scores of 0.70 for speech-speech and 0.71 for speech-text tasks. These results established a benchmark for

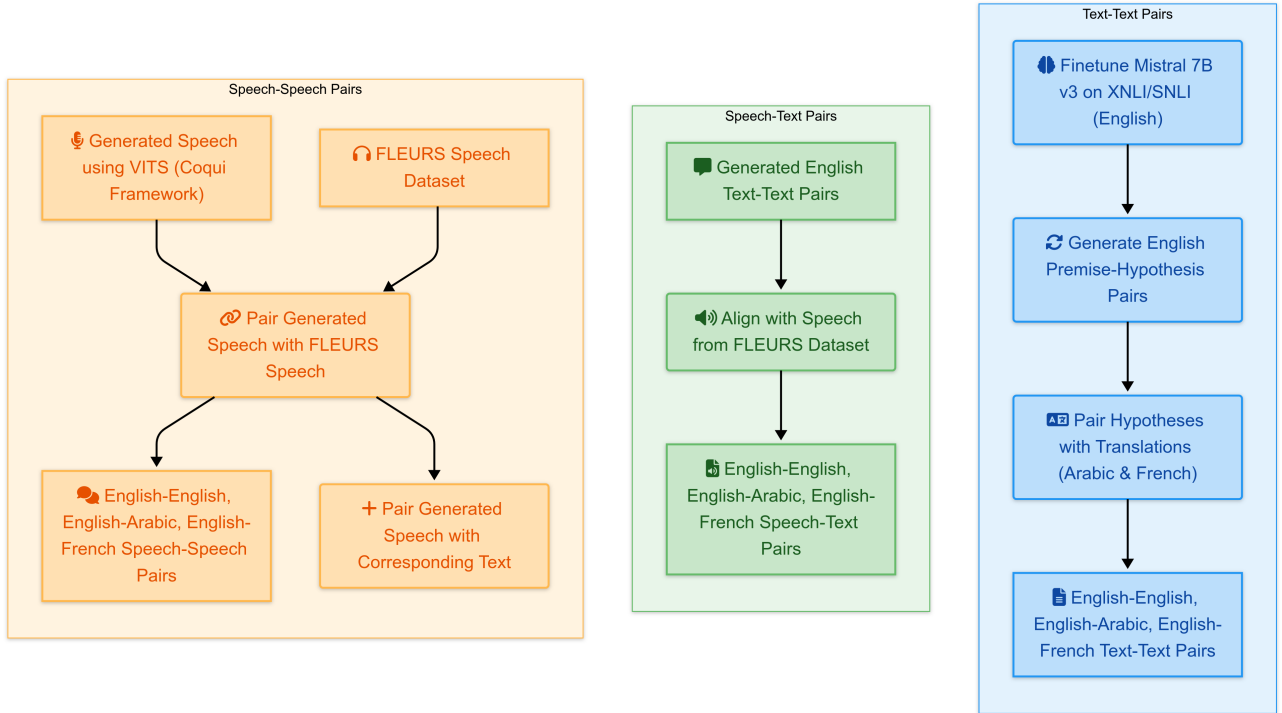


Figure 2: Overview of the data generation process. This includes text-text pairs generated using the Mistral 7B model, speech-text pairs aligned with the FLEURS dataset, and speech-speech pairs created using VITS and paired with FLEURS speech.

Table 3: Comparison by Example: BLASER vs. Our Model

Premise	Hypothesis	Hypothesis Translation	BLASER	Our Model
I love chocolate.	J’adore le chocolat.	I love chocolate.	4.39	Entailment
I love chocolate.	Je déteste le chocolat.	I hate chocolate.	4.49	Contradiction

Table 4: Test Set Performance: Impact of Generated and Synthesized Data (F1 Micro). The Baseline includes XNLI-SNLI. The + GT adds Generated Text (text-text and speech-text pairs), while + GT + GTS adds both Generated Text and Speech

Test Split	Baseline	+ GT	+ GT + GTS
speech-speech	0.70	0.70	0.77
speech-text	0.71	0.74	0.76

evaluating the effect of adding generated and synthesized data.

In the second experiment, we introduced generated text paired with natural speech into the XNLI-SNLI training data. This improved F1 Micro scores to 0.70 for speech-speech and 0.74 for speech-text tasks, demonstrating that generated text paired with natural speech enhances speech-text classification.

The third experiment included generated speech alongside generated text and natural speech. The generated text was paired with both generated and natural speech, while the generated speech was paired with natural speech. This setup yielded the highest scores, with 0.77 for speech-speech and 0.76 for speech-text, emphasizing the benefits of combining generated text and speech with natural data. Table 4 summarizes these results, showing that generated text paired with natural speech improves performance while adding generated speech further enhances it.

4. Conclusion

This paper introduced a multilingual and multimodal NLI framework that classifies relationships across text-text, speech-text, and speech-speech pairs in both same-language and cross-lingual settings. Using shared semantic representations, the framework models entailment, contradiction, and neutrality, overcoming the limitations of similarity-based models.

Experiments demonstrated that the proposed model outperforms BLASER in distinguishing entailment from non-entailment, with notable improvements of 0.19 F1 in speech-speech classification and 0.13 F1 in speech-text classification. Multiclass experiments highlighted the effectiveness of combining generated and natural data for improving multimodal NLI tasks. The framework identifies contradictions, offering additional insights for translation and speech evaluation alongside semantic similarity in retrieval and assessment. Additionally, the generated dataset supports ongoing research in multimodal NLI and cross-lingual inference.

However, performance on speech-text (F1 = 0.76) and speech-speech (F1 = 0.77) in the three-class NLI classification indicates room for improvement. While text-based evaluation was conducted on real data, speech-text, and speech-speech tasks relied primarily on generated text and synthesized speech. Further validation of real-world spoken data would strengthen the model’s applicability.

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6. References

- [1] S. Bowman, G. Angeli, C. Potts, and C. D. Manning, “A large annotated corpus for learning natural language inference,” in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, 2015, pp. 632–642.
- [2] B. MacCartney, *Natural language inference*. Stanford University, 2009.
- [3] R. Gubelmann, I. Katis, C. Niklaus, and S. Handschuh, “Capturing the varieties of natural language inference: A systematic survey of existing datasets and two novel benchmarks,” *Journal of Logic, Language and Information*, vol. 33, no. 1, pp. 21–48, 2024.
- [4] Y. Nie, A. Williams, E. Dinan, M. Bansal, J. Weston, and D. Kiela, “Adversarial nli: A new benchmark for natural language understanding,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 4885–4901.
- [5] R. Khobragade, H. Patel, A. Namdev, A. Mishra, and P. Bhat-tacharyya, “Machine translation evaluation using bi-directional entailment,” *arXiv preprint arXiv:1911.00681*, 2019.
- [6] B. MacCartney and C. D. Manning, “An extended model of natural logic,” in *Proceedings of the eight international conference on computational semantics*, 2009, pp. 140–156.
- [7] T. Rocktäschel, E. Grefenstette, K. M. Hermann, T. Kočiský, and P. Blunsom, “Reasoning about entailment with neural attention,” *arXiv preprint arXiv:1509.06664*, 2015.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 4171–4186.
- [9] Y. Liu, “Roberta: A robustly optimized bert pretraining approach,” *arXiv preprint arXiv:1907.11692*, vol. 364, 2019.
- [10] L. Xue, “mt5: A massively multilingual pre-trained text-to-text transformer,” *arXiv preprint arXiv:2010.11934*, 2020.
- [11] O. Shliazhko, A. Fenogenova, M. Tikhonova, V. Mikhailov, A. Kozlova, and T. Shavrina, “mgpt: Few-shot learners go multilingual,” *arXiv preprint arXiv:2204.07580*, 2022.
- [12] M. R. Costa-jussà, J. Cross, O. Çelebi, M. Elbayad, K. Heffernan, E. Kalbassi, J. Lam, D. Licht, J. Maillard *et al.*, “No language left behind: Scaling human-centered machine translation,” *arXiv preprint arXiv:2207.04672*, 2022.
- [13] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, “Language-agnostic bert sentence embedding,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 878–891.
- [14] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *Advances in neural information processing systems*, vol. 33, pp. 12 449–12 460, 2020.
- [15] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhota, R. Salakhutdinov, and A. Mohamed, “Hubert: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 3451–3460, 2021.
- [16] P.-A. Duquenne, H. Schwenk, and B. Sagot, “Sonar: sentence-level multimodal and language-agnostic representations,” *arXiv e-prints*, pp. arXiv–2308, 2023.
- [17] S. Khurana, A. Laurent, and J. Glass, “Samu-xlsr: Semantically-aligned multimodal utterance-level cross-lingual speech representation,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1493–1504, 2022.
- [18] D. Cer, Y. Yang, S.-y. Kong, N. Hua, N. Limtiaco, R. S. John, N. Constant, M. Guajardo-Cespedes, S. Yuan, C. Tar *et al.*, “Universal sentence encoder for english,” in *Proceedings of the 2018 conference on empirical methods in natural language processing: system demonstrations*, 2018, pp. 169–174.
- [19] N. Reimers, “Sentence-bert: Sentence embeddings using siamese bert-networks,” *arXiv preprint arXiv:1908.10084*, 2019.
- [20] B. Zhang and C. Li, “Advancing semantic textual similarity modeling: A regression framework with translated relu and smooth k2 loss,” *arXiv preprint arXiv:2406.05326*, 2024.
- [21] O. Nacar and A. Koubaa, “Enhancing semantic similarity understanding in arabic nlp with nested embedding learning,” *arXiv preprint arXiv:2407.21139*, 2024.
- [22] L. Barrault, Y.-A. Chung, M. C. Meglioli, D. Dale, N. Dong, P.-A. Duquenne, H. Elsahar, H. Gong, K. Heffernan, J. Hoffman *et al.*, “Seamlessm4t—massively multilingual & multimodal machine translation,” *ArXiv*, 2023.
- [23] M. Chen, P.-A. Duquenne, P. Andrews, J. Kao, A. Mourachko, H. Schwenk, and M. R. Costa-jussà, “Blaser: A text-free speech-to-speech translation evaluation metric,” *arXiv preprint arXiv:2212.08486*, 2022.
- [24] D. Licht, C. Gao, J. Lam, F. Guzmán, M. Diab, and P. Koehn, “Consistent human evaluation of machine translation across language pairs,” in *Proceedings of the 15th biennial conference of the Association for Machine Translation in the Americas (Volume 1: Research Track)*, 2022, pp. 309–321.
- [25] A. Conneau, R. Rinott, G. Lample, A. Williams, S. Bowman, H. Schwenk, and V. Stoyanov, “Xnli: Evaluating cross-lingual sentence representations,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 2475–2485.
- [26] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, “Fleurs: Few-shot learning evaluation of universal representations of speech,” in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 798–805.
- [27] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. d. l. Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier *et al.*, “Mistral 7b,” *arXiv preprint arXiv:2310.06825*, 2023.
- [28] J. Kim, J. Kong, and J. Son, “Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 5530–5540.
- [29] E. Gölge and T. C. T. Team, “Coqui tts,” 2021. [Online]. Available: <https://www.coqui.ai>
- [30] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *International conference on machine learning*. PMLR, 2023, pp. 28 492–28 518.